

A Practical Guide to Protein structure prediction

TOPIC -- PROTEIN STRUCTURE PREDICTION

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The α -helix is the most abundant type of secondary structure in proteins. The α -helix has 3.6 amino acids per turn with an H-bond formed between every fourth residue; the average length is 10 amino acids (3 turns) or 10 Å but varies from 5 to 40 (1.5 to 11 turns). The alignment of the H-bonds creates a dipole moment for the helix with a resulting partial positive charge at the amino end of the helix. Because this region has free NH₂ groups, it will interact with negatively charged groups such as phosphates. The most common location of α -helices is at the surface of protein cores, where they provide an interface with the aqueous environment. The inner-facing side of the helix tends to have hydrophobic amino acids and the outer-facing side hydrophilic amino acids. Thus, every third or fourth amino acids along the chain will tend to be hydrophobic, a pattern that can be quite readily detected. In the leucine zipper motif, a repeating pattern of leucines on the facing sides of two adjacent helices is highly predictive of the motif. A helical-wheel plot can be used to show this repeated pattern. Other α -helices buried in the protein core or in cellular membranes have a higher and more regular distribution of hydrophobic amino acids, and are highly predictive of such structures. Helices exposed on the surface have a lower proportion of hydrophobic amino acids. Amino acid content can be predictive of an α -helical region. Regions richer in alanine (A), glutamic acid (E), leucine (L), and methionine (M) and poorer in proline (P), glycine (G), tyrosine (Y), and serine (S) tend to form an α -helix. Proline destabilizes or breaks an α -helix but can be present in longer helices, forming a bend.

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Background Early methods of secondary structure prediction, introduced in the 1960s and early 1970s, focused on identifying likely alpha helices and were based mainly on helix-coil transition models. Significantly more accurate predictions that included beta sheets were introduced in the 1970s and relied on statistical assessments based on probability parameters derived from known solved structures. These methods, applied to a single sequence, are typically at most about 60–65% accurate, and often underpredict beta sheets. Since the 1980s, artificial neural networks have been applied to the prediction of protein structures. The evolutionary conservation of secondary structures can be exploited by simultaneously assessing many homologous sequences in a

multiple sequence alignment, by calculating the net secondary structure propensity of an aligned column of amino acids. In concert with larger databases of known protein structures and modern machine learning methods such as neural nets and support vector machines, these methods can achieve up to 80% overall accuracy in globular proteins. The theoretical upper limit of accuracy is around 90%, partly due to idiosyncrasies in DSSP assignment near the ends of secondary structures, where local conformations vary under native conditions but may be forced to assume a single conformation in crystals due to packing constraints. Moreover, the typical secondary structure prediction methods do not account for the influence of tertiary structure on formation of secondary structure; for example, a sequence predicted as a likely helix may still be able to adopt a beta-strand conformation if it is located within a beta-sheet region of the protein and its side chains pack well with their neighbors. Dramatic conformational changes related to the protein's function or environment can also alter local secondary structure.

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The practical role of protein structure prediction is now more important than ever. Massive amounts of protein sequence data are produced by modern large-scale DNA sequencing efforts such as the Human Genome Project. Despite community-wide efforts in structural genomics, the output of experimentally determined protein structures—typically by time-consuming and relatively expensive X-ray crystallography or NMR spectroscopy—is lagging far behind the output of protein sequences. The protein structure prediction remains an extremely difficult and unresolved undertaking. The two main problems are the calculation of protein free energy and finding the global minimum of this energy. A protein structure prediction method must explore the space of possible protein structures which is astronomically large. These problems can be partially bypassed in "comparative" or homology modeling and fold recognition methods, in which the search space is pruned by the assumption that the protein in question adopts a structure that is close to the experimentally determined structure of another homologous protein. In contrast, the de novo protein structure prediction methods must explicitly resolve these problems. The progress and challenges in protein structure prediction have been reviewed by Zhang.

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The peptide bonds in the chain are polar, i.e. they have separated positive and negative charges (partial charges) in the carbonyl group, which can act as hydrogen bond acceptor and in the NH group, which can act as hydrogen bond

donor. These groups can therefore interact in the protein structure. Proteins consist mostly of 20 different types of L- α -amino acids (the proteinogenic amino acids). These can be classified according to the chemistry of the side chain, which also plays an important structural role. Glycine takes on a special position, as it has the smallest side chain, only one hydrogen atom, and therefore can increase the local flexibility in the protein structure. Cysteine in contrast can react with another cysteine residue to form one cystine and thereby form a cross link stabilizing the whole structure. Protein structure arises from a sequence of secondary structure elements, such as α helices and β sheets. In secondary structures, regular patterns of H-bonds are formed between the main chain NH and CO groups of spatially neighboring amino acids, and the amino acids have similar Φ and ψ angles.

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Protein classification Proteins may be classified according to both structural and sequential similarity. For structural classification, the sizes and spatial arrangements of secondary structures described in the above paragraph are compared in known three-dimensional structures. Classification based on sequence similarity was historically the first to be used. Initially, similarity based on alignments of whole sequences was performed. Later, proteins were classified on the basis of the occurrence of conserved amino acid patterns. Databases that classify proteins by one or more of these schemes are available. In considering protein classification schemes, it is important to keep several observations in mind. First, two entirely different protein sequences from different evolutionary origins may fold into a similar structure. Conversely, the sequence of an ancient gene for a given structure may have diverged considerably in different species while at the same time maintaining the same basic structural features. Recognizing any remaining sequence similarity in such cases may be a very difficult task. Second, two proteins that share a significant degree of sequence similarity either with each other or with a third sequence also share an evolutionary origin and should share some structural features also. However, gene duplication and genetic rearrangements during evolution may give rise to new gene copies, which can then evolve into proteins with new function and structure.